

CLI Usage

CLI Usage

1. Using CLI for Prediction
 2. CLI Usage Syntax
 3. Image Prediction
 - Effect Preview
 4. Video Prediction
 5. Real-time Prediction
- Reference Materials

1. Using CLI for Prediction

CLI (Command-Line Interface) is a way for users to interact with computers or software programs, where users run programs or execute tasks by typing text commands, rather than clicking icons or buttons through a graphical user interface (GUI).

2. CLI Usage Syntax

```
yolo TASK MODE ARGS
```

where TASK (optional) is one of [detect, segment, classify, pose, obb]
MODE (required) is one of [train, val, predict, export, track, benchmark]
ARGS (optional) are any number of custom 'arg=value' pairs like 'imgsz=320' that override defaults.

3. Image Prediction

Use yolo26n.pt to predict the built-in images in the ultralytics26 project: If the system cannot find the corresponding model file in the directory where the command is run, it will automatically download (if download fails, you can copy the model in yourself)

Enter the project folder:

```
cd ~/ultralytics/
```

Use yolo26n.pt to detect images in the target folder and output results:

```
yolo predict model=yolo26n.pt source='assets/bus.jpg'
```

```
root@jetson-yahboom:/ultralytics# yolo predict model=yolo26n.pt source='/root/yolo26_data/bus.jpg'
Ultralytics 8.4.24 Python-3.8.0 torch-1.11.0a0+gitbc2c6ed CUDA:0 (NVIDIA Tegra X1, 3964MiB)
/ultralytics/ultralytics/nn/modules/head.py:246: UserWarning: __floordiv__ is deprecated, and its behavior will change in a future version of pytorch. It currently rounds toward 0 (like the 'trunc' function NOT 'floor'). This results in incorrect rounding for negative values. To keep the current behavior, use torch.div(a, b, rounding_mode='trunc'), or for actual floor division, use torch.div(a, b, rounding_mode='floor').
  idx = ori_index[torch.arange(batch_size)[..., None], index // nc] # original index
YOLO26n summary (fused): 122 layers, 2,408,932 parameters, 0 gradients, 5.4 GFLOPs

image 1/1 /root/yolo26_data/bus.jpg: 640x480 4 persons, 1 bus, 130.9ms
Speed: 14.6ms preprocess, 130.9ms inference, 2.4ms postprocess per image at shape (1, 3, 640, 480)
Results saved to /ultralytics/runs/detect/predict2
? Learn more at https://docs.ultralytics.com/modes/predict
```

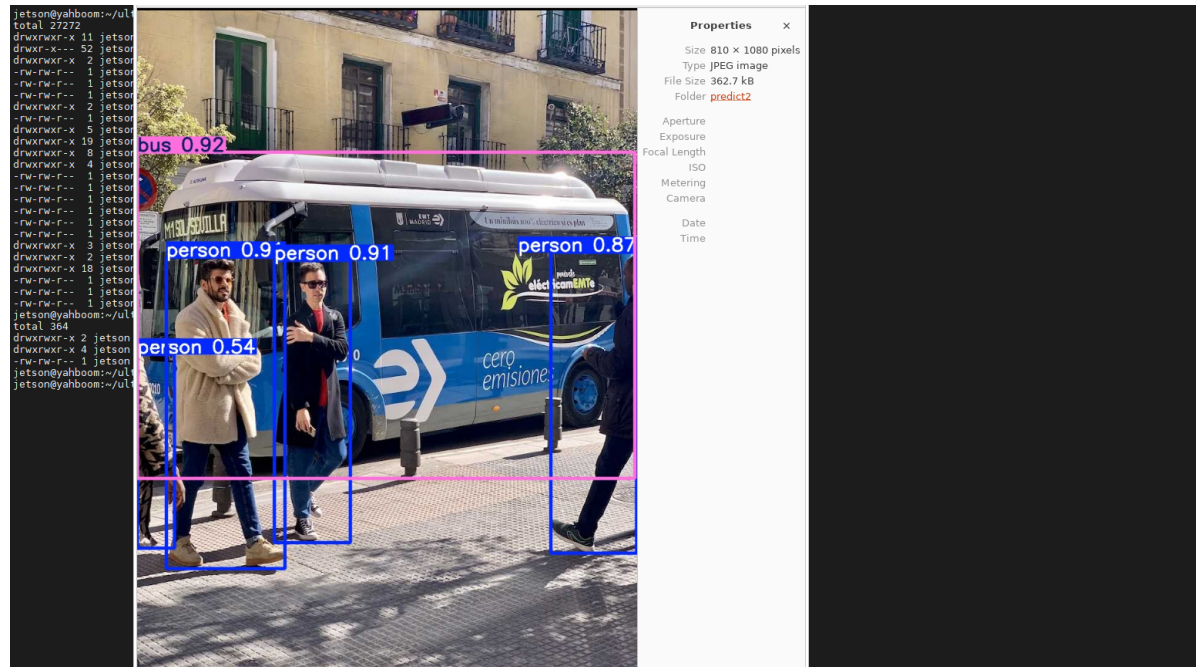
Effect Preview

yolo recognition output image location: /home/jetson/ultralytics/runs/detect/predict2

Each time it runs, the predict under this folder automatically increments by 1

```
jetson@yahboom:~/ultralytics$ yolo predict model=yolo26n.pt source=assets/bus.jpg
Ultralytics 8.4.24 Python-3.10.12 torch-2.5.0a0-872d972e41.nv24.08 CUDA:0 (0rtn, 7620MiB)
YOL026n summary (fused): 122 layers, 2,408,932 parameters, 0 gradients, 5.4 GFLOPs

image 1/1 /home/jetson/ultralytics/assets/bus.jpg: 640x480 4 persons, 1 bus, 174.2ms
Speed: 10.0ms preprocess, 174.2ms inference, 16.0ms postprocess per image at shape (1, 3, 640, 480)
Results saved to /home/jetson/ultralytics/runs/detect/predict2
Learn more at https://docs.ultralytics.com/modes/predict
jetson@yahboom:~/ultralytics$
```



4. Video Prediction

Use yolo26n.pt to predict videos in the ultralytics project (not built-in ultralytics videos): If the system cannot find the corresponding model file in the directory where the command is run, it will automatically download (if download fails, you can copy the model in yourself)

Enter the project folder:

```
cd ~/ultralytics
```

Use yolo26n.pt to detect the target video and output results:

Here we switched to yolo26n.engine, which is the TensorRT model we converted in the previous section. The performance on Jetson is better than yolo26n.pt

```
yolo predict model=yolo26n.engine
source='/home/jetson/ultralytics/videos/car.mp4'
```

[illegible]

Alternatively, we can directly add the show parameter to watch the video

```
yolo predict model=yolo26n.engine
source='/home/jetson/ultralalytics/videos/car.mp4' show
```

`python3 yolov5.py --source=/home/jetson/ulalytics/videos/car.mp4`

WARNING: Unable to automatically guess model task, assuming 'task=detect'. Explicitly define task for your model, i.e. 'task=detect', 'segment', 'classify', 'pose' or 'obb'

Ultralytics 8.4.24 • Python-3.10.12 torch-2.5.0ad872d972e41.nv24.08 CUDA:0 (0rn, 7620MiB)

Loading yolo26n engine for TensorRT inference.

[03/21/2026
[03/21/2026
[03/21/2026

X /home/jetson/ulalytics/videos/car.mp4@yahboom

The image displays a real-time video feed of a busy multi-lane highway. Numerous vehicles, primarily cars, are visible in both directions of travel. A computer vision algorithm, YOLOv5, has been applied to the video, resulting in white rectangular bounding boxes around each detected vehicle. Each box is accompanied by a label 'car' and a numerical confidence score ranging from approximately 0.26 to 0.85. On the right side of the road, a traffic light pole is visible, also identified by the system with a label 'traffic light 0.35'. The video player window includes a terminal-like output area on the left showing command-line information, warnings about automatic task detection, Ultralytics version details, and loading status for TensorRT inference. The top of the window shows standard OS window controls and the file path of the video being played.

5. Real-time Prediction

Use yolo26n.pt to predict USB camera footage: If the system cannot find the corresponding model file in the directory where the command is run, it will automatically download (if download fails, you can copy the model in yourself)

Note: CLI method does not support CSI cameras

Enter the project folder:

```
cd ~/ultralytics
```


Use yolo26n.pt to detect camera footage and output results: Currently only USB cameras can directly use CLI to predict real-time footage. No relevant information found for CSI cameras as direct input sources

```
yolo predict model=yolo26n.engine source=0 save=False show # Object detection
# Instance segmentation: yolo predict model=yolo26n-seg.pt source=0 save=False
show
# Image classification: yolo predict model=yolo26n-cls.pt source=0 save=False
show
# Pose estimation: yolo predict model=yolo26n-pose.pt source=0 save=False show
# Oriented object detection: yolo predict model=yolo26n-obb.pt source=0
save=False show
```

Click the terminal and press "Ctrl + C" shortcut to terminate the program!

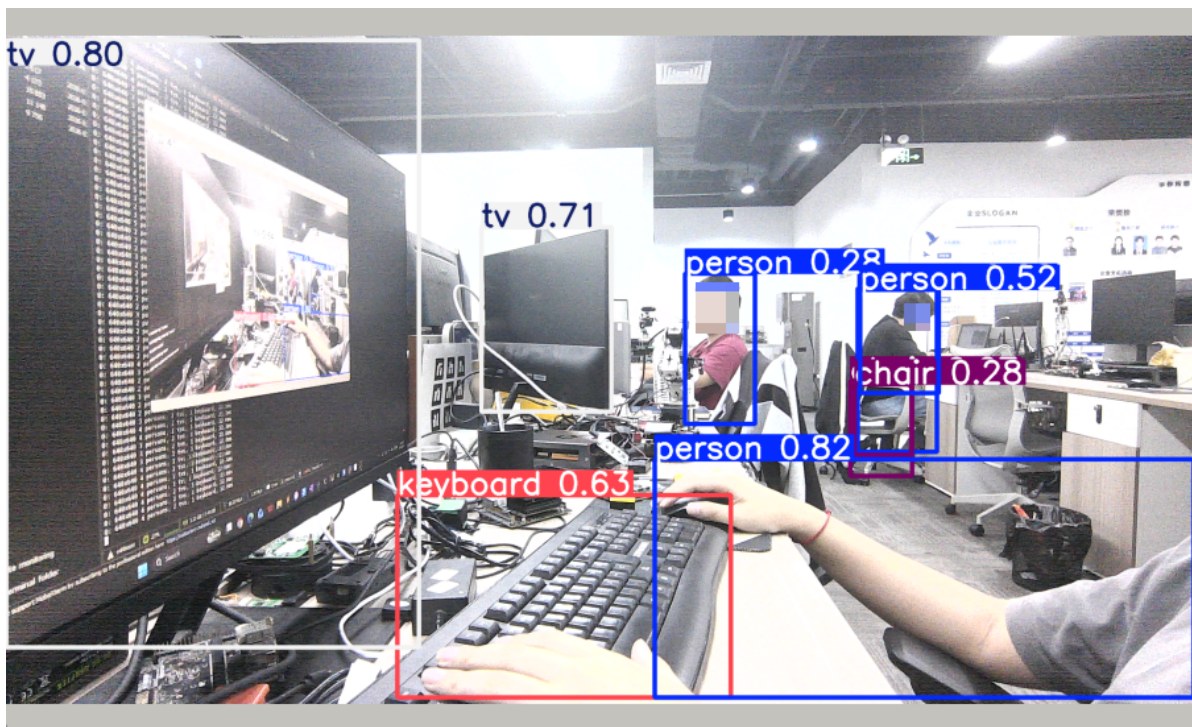
Parameter Description

model: Specify YOLO model

source: Specify recognition source: multiple cameras can switch numbers

save=False: Disable saving results

show: Real-time display



Reference Materials

[Home - Ultralytics YOLO Documentation](#)

[Command Line Interface - Ultralytics YOLO Documentation](#)